

## USE-CASE FLOW

### 1. Preconditions

1. The microservice is registered in the service catalog.
2. A health endpoint is available for the microservice.
3. The health-check monitoring process is active.
4. The DevOps Engineer has access to the portal.

### 2. Main Success Scenario

1. The system starts the health-check cycle.
2. The system sends a health probe to the microservice health endpoint.
3. The microservice responds successfully to the health probe.
4. The system records the successful health-check result.
5. The system updates the service availability information.
6. The system updates the rolling 24-hour service availability percentage.
7. The system waits 30 seconds before the next health probe.
8. The DevOps Engineer can view the current health status of the microservice.

### 3. Alternate Flow

- 3a. Microservice does not respond to the health probe.
  - 3a1. The system records the failed health probe.
  - 3a2. The system waits 30 seconds and sends the next health probe.
  - 3a3. If three consecutive health probes fail, the system records the microservice as down.
  - 3a4. The system records the downtime event.
  - 3a5. The system dispatches a downtime alert to the DevOps Engineer.
  - 3a6. The DevOps Engineer investigates the affected microservice.

### 4. Postconditions

1. The latest health-check result is recorded.
2. The rolling 24-hour availability percentage is updated.
3. If three consecutive probes fail, downtime is recorded and an alert is dispatched.